

# E-COMMERCE SALES ANALYSIS PROJECT

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## Project Overview

This project is a **complete data analysis workflow** built using Python. It analyzes an e-commerce sales dataset to extract insights such as:

- Total sales performance
- Average sales per transaction
- Best-selling product
- Sales trends over time
- Product-wise contribution to revenue

The project includes:

- ✓ Data loading
- ✓ Data cleaning
- ✓ Data validation
- ✓ Statistical analysis
- ✓ Data visualization
- ✓ Error handling
- ✓ Insight generation

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## Project Objective

The main goal of this project is to:

- Analyze e-commerce sales data
  - Identify the best-performing product
  - Understand sales distribution
  - Visualize trends using charts
  - Build a professional data analysis pipeline
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# Tools & Technologies Used

- Python 3
  - Pandas (Data manipulation & analysis)
  - Matplotlib (Data visualization)
  - OS module (File handling)
  - GitHub (Version control)
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## Project Structure

E-Commerce Sales Analysis/

```
|
|— README.md
|— main.py
|— requirements.txt
|
|— data/
|   |— sales_data.csv
|
|— visualizations/
|   |— bar_sales_by_product.png
|   |— line_sales_trend.png
|   |— pie_sales_distribution.png
|
|— report/
|   |— report.md
```

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## Dataset Description

The dataset used in this project is a simple e-commerce sales dataset.

### Required Columns:

| Column Name | Description |
|-------------|-------------|
|-------------|-------------|

|             |                                   |
|-------------|-----------------------------------|
| Product     | Name of the product               |
| Total_Sales | Sales amount for each transaction |

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## Features of the Project

- ✓ Loads dataset using Pandas
  - ✓ Handles missing values
  - ✓ Removes duplicate records
  - ✓ Validates required columns
  - ✓ Calculates key metrics
  - ✓ Finds best-selling product
  - ✓ Generates visualizations
  - ✓ Saves charts automatically
  - ✓ Includes error handling
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## Step-by-Step Workflow

### 1. Data Loading

- Dataset is loaded using `pandas.read_csv()`

### 2. Data Exploration

- Shape of dataset
- Column names
- Missing values check

### 3. Data Cleaning

- Missing values filled with 0
- Duplicate rows removed

### 4. Data Validation

- Ensures required columns exist:
  - Product
  - Total\_Sales

## 5. Data Analysis

- Total Sales calculation
  - Average Sales calculation
  - Grouping sales by product
  - Identifying best-selling product
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# Visualizations Created

## 1. Bar Chart – Sales by Product

- Compares total sales across products
- Helps identify top-performing products

Saved as:

visualizations/bar\_sales\_by\_product.png

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## 2. Line Chart – Sales Trend

- Shows sales pattern across transactions
- Helps identify fluctuations in sales

Saved as:

visualizations/line\_sales\_trend.png

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## 3. Pie Chart – Product Contribution

- Shows percentage contribution of each product
- Helps understand revenue distribution

Saved as:

visualizations/pie\_sales\_distribution.png

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## Key Outputs

When you run the project, you will see:

### Console Output:

- Dataset shape
- Missing values
- Total sales
- Average sales
- Best-selling product
- Insights summary

### Generated Charts:

- Bar chart
  - Line chart
  - Pie chart
- 

## How to Run This Project

### Step 1: Install Requirements

```
pip install pandas matplotlib
```

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### Step 2: Add Dataset

Make sure `sales_data.csv` is in the same folder as `main.py`

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### Step 3: Run the Program

```
python main.py
```

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## Error Handling Included

This project handles:

- ✓ File not found error
- ✓ Missing column error
- ✓ Unexpected runtime errors

Example:

- If dataset is missing → shows clear error message
- If columns are incorrect → stops execution safely

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## Insights from Analysis

- Product-wise performance comparison helps identify top products
- Sales trend chart shows how sales vary across transactions
- Pie chart shows contribution of each product to total revenue
- Best-selling product is automatically identified using grouping logic

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## Learning Outcomes

After completing this project, you will learn:

- How to perform real-world data analysis
  - Data cleaning techniques in Pandas
  - Data grouping and aggregation
  - Creating professional charts using Matplotlib
  - Error handling in Python projects
  - Structuring a GitHub-ready project
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# Output Files Generated

After execution:

## visualizations/

- bar\_sales\_by\_product.png
  - line\_sales\_trend.png
  - pie\_sales\_distribution.png
- 

## Future Improvements

- Add interactive dashboard (Streamlit)
  - Use real-time datasets
  - Add machine learning prediction
  - Improve UI for reports
  - Export PDF reports automatically
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## Conclusion

This project successfully demonstrates a complete data analysis workflow using Python. Starting from raw sales data, the project performs data cleaning, validation, analysis, and visualization to extract meaningful business insights.

The analysis highlights product performance, overall sales trends, and revenue contribution through clear and simple visual charts. The generated graphs make it easier to understand patterns in the data and support data-driven decision making.

Overall, this project helped in strengthening key skills such as data manipulation with Pandas, visualization using Matplotlib, and structured problem-solving in a real-world dataset scenario.

## E-COMMERCE SALES ANALYSIS

Dataset Loaded Successfully!

Dataset Shape: (100, 7)

Columns: ['Date', 'Product', 'Quantity', 'Price', 'Customer\_ID', 'Region', 'Total\_Sales']

Missing Values:

|             |       |
|-------------|-------|
| Date        | 0     |
| Product     | 0     |
| Quantity    | 0     |
| Price       | 0     |
| Customer_ID | 0     |
| Region      | 0     |
| Total_Sales | 0     |
| dtype:      | int64 |

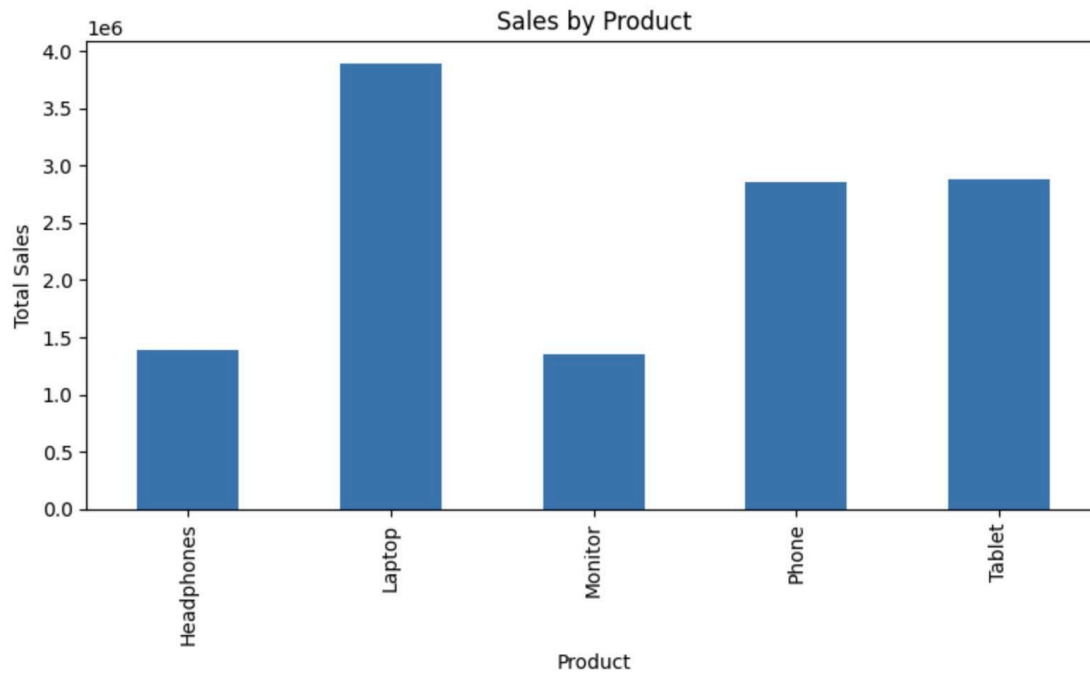
Data Cleaning Completed!

## SALES REPORT

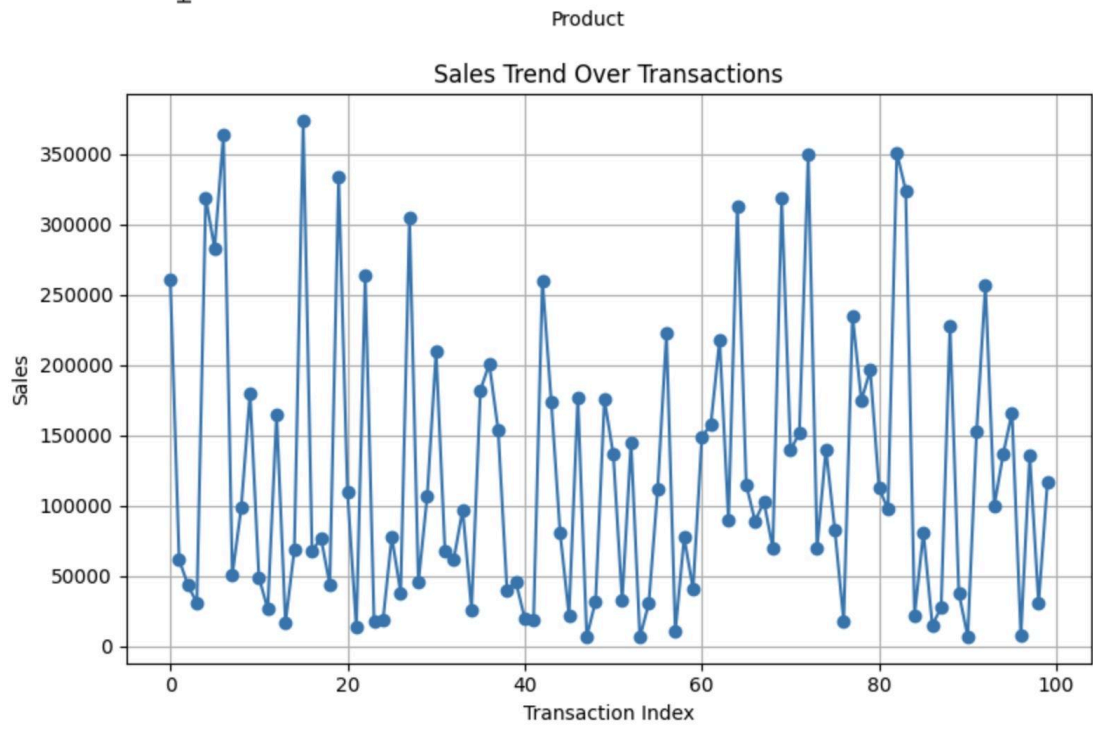
Total Sales : ₹12,365,048.00

Average Sales: ₹123,650.48

Best Product : Laptop

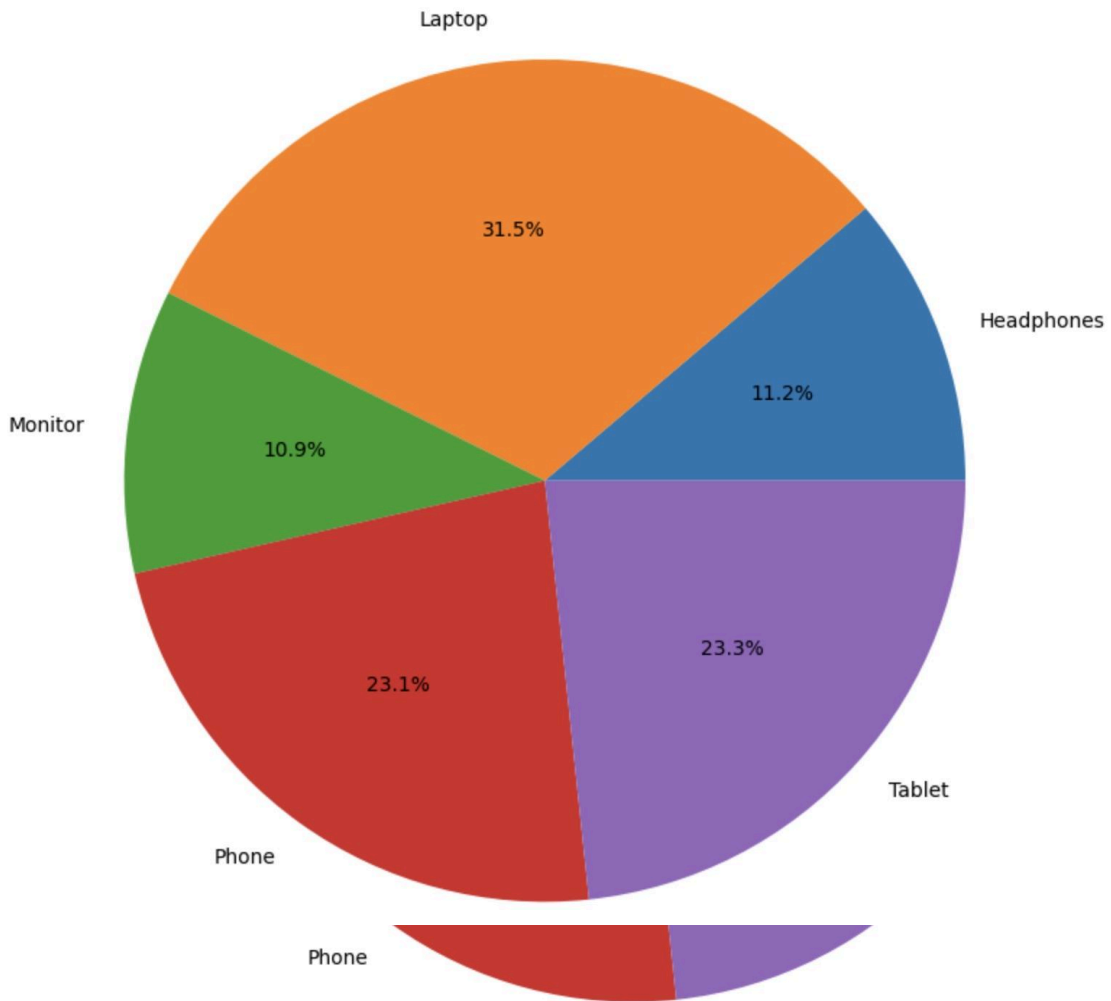






Product-wise Sales Distribution

## Product-wise Sales Distribution



### INSIGHTS

1. Bar chart compares product performance.
2. Line chart shows sales fluctuations over time.
3. Pie chart shows contribution of each product.
4. Best selling product: Laptop

PROJECT COMPLETED SUCCESSFULLY!